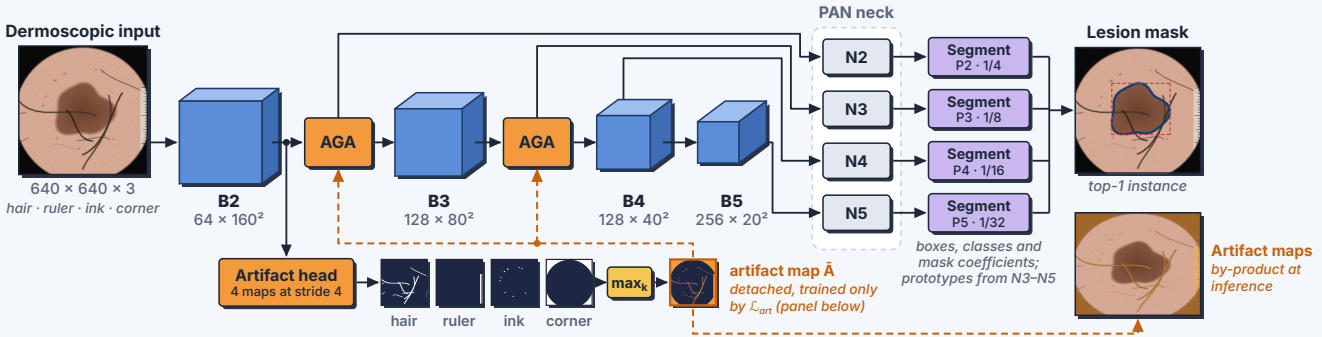
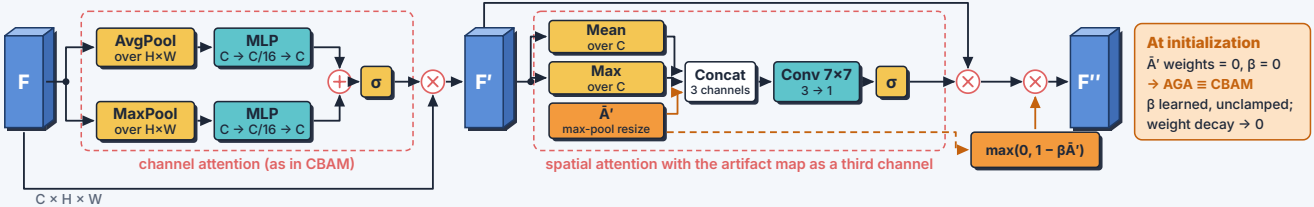
# ArtiGate-YOLO

*YOLO26n-seg with a P2 level · AGA replaces CBAM after B2 and B3*




# AGA

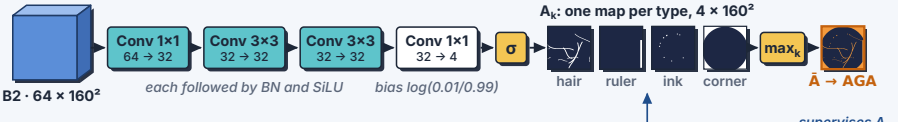
*artifact-gated attention · drop-in replacement for CBAM*




$$M_c = \sigma(\mathrm{MLP}(\mathrm{Avg}_{HW}\,F) + \mathrm{MLP}(\mathrm{Max}_{HW}\,F)),\ F' = M_c \odot F \qquad M_s = \sigma(\mathrm{Conv}_{7\times7}[\mathrm{Mean}_C\,F';\ \mathrm{Max}_C\,F';\ \bar{A}']),\ F'' = M_s \odot F' \odot \max(0,\ 1 - \beta\bar{A}')$$

# Artifact head

*trained without manual masks: synthetic artifacts and partial labels*




**Targets per type and pixel**

- **present** synthetic artifact masks
- **absent** image-level flags, colour/morphology rules
- **unknown** everything else, ignored by the loss

$\mathcal{L}_{art}$: BCE + soft Dice on labelled cells, averaged over types; $\mathcal{L} = \mathcal{L}_{seg} + \lambda\,\mathcal{L}_{art}$

Conv · Pooling / σ / max · Segment26 head · Neck · Added by ArtiGate-YOLO · Feature map · Artifact map Ā · Element-wise product · Sum